ABDELRAHMAN ELGRM

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EXECUTIVE SUMMARY

Data Scientist and Python developer with hands-on experience building end-to-end ML/DL pipelines, computer vision systems, and NLP solutions using TensorFlow, PyTorch, and Scikit-learn. Currently pursuing a degree in Electrical and Computer Engineering, seeking an AI/Data Science internship to solve real-world problems with intelligent, scalable solutions.

EDUCATION

Menofia University Oct. 2022 – Jan. 2027
Bachelor in Electrical and Computer Engineering *GPA: 3.47*

PROFESSIONAL EXPERIENCE

ELARABY GROUP Jul. 2025 – Sept. 2025
Engineering Intern

- **Designed** a cross-platform Python/PyQt application for factory automation and developed a Computer Vision model using TensorFlow to detect adhesive precision on LED TV backlights, reducing defect escape rates by **25%**.
- Executed loss prevention framework checks and conducted comprehensive technical audits to streamline assembly workflows, increasing operational throughput by **15%**.

ALX Africa Jan. 2025 – Jun. 2025
Data Scientist Intern

- Constructed end-to-end ML/DL, Computer Vision, and NLP solutions using TensorFlow and PyTorch alongside **automated** Python/SQL pipelines and Power BI dashboards.
- Managed full data lifecycles from preprocessing to deployment, transforming complex industrial datasets into actionable KPIs for key stakeholders.

TECHNICAL PROJECTS

NEURASHIELD AI-SOC Analyst | ***Python, FastAPI, TensorFlow, Supabase, LLMs, RAG, SQL***

- Developed an autonomous SOC agent using **FastAPI** and **Supabase** to ingest raw security logs, integrating a **RAG pipeline** to contextualize anomalies against cybersecurity frameworks.
- Applied fine-tuned LLMs to auto-generate incident response playbooks and utilized **TensorFlow** with vector search to accelerate intelligence retrieval times by **35%.**

UCI Heart Disease Predictor | ***Python, Scikit-learn, XGBoost, Pandas, Streamlit***

- Built an end-to-end clinical ML pipeline on **920 patient records** from 4 hospitals, tuning Logistic Regression, Random Forest, and XGBoost with **GridSearchCV**; the selected Random Forest reached **96% recall** and **0.934 ROC-AUC**, missing only 4 of 102 sick patients.

Premier League Match Predictor | ***Python, XGBoost, Scikit-learn, FastAPI, Streamlit***

- Engineered **27 leak-free features** (Elo ratings, rolling form, league position, head-to-head) from **9,000+ matches** since 2000; with a strict time-based split, XGBoost reached **53.4% accuracy** vs. a 43.2% home-win baseline.

Feasibility Study Desktop App | ***Python, PyQt6, QThread***

- **Engineered** a multithreaded desktop application for industrial feasibility studies, deploying **QThread** to maintain 100% UI responsiveness during intensive calculations.
- Designed an intuitive **PyQt6** interface that streamlined manual data entry and **automated** report generation for engineering workflows.

Computer Vision for Manufacturing | ***OpenCV, TensorFlow, CNNs***

- Built a dual-purpose **CNN system** to optimize assembly line quality control, substantially lowering manual inspection costs.
- Enforced real-time object tracking algorithms with **OpenCV** for continuous security and process monitoring across manufacturing facilities.

Maji Ndogo Water Crisis Analysis | ***SQL, Python, Power BI***

- Analyzed **60,000+ records** using SQL queries to isolate critical infrastructure failures across regional water distribution networks.
- Built interactive **Power BI dashboards** to visualize resource allocation gaps and guide data-driven strategy for key stakeholders.

Movie Recommendation System | ***Python, Scikit-learn, Pandas, Flask***

- Architected a hybrid collaborative and content-based recommendation engine using **Scikit-learn**, delivering top-tier accuracy on user preference datasets.
- Deployed model predictions via a **Flask REST API**, supporting dynamic real-time recommendation endpoints.

COURSES & CERTIFICATIONS

ML Specialization (DeepLearning.AI): Supervised ML (Regression, Classification, Gradient Descent), Advanced Algorithms (MLPs, Ensemble Models, TensorFlow), and Unsupervised Learning (K-Means, PCA, Deep Q-Networks).

CS109: Probability for Computer Scientists (Stanford): Statistical inference, probabilistic modeling, and stochastic analysis.

Machine Learning Engineer Track (DataCamp): End-to-end ML engineering, covering model development, deployment, and production pipelines.

SKILLS & LANGUAGES

Technical Skills: Python · SQL · TensorFlow · PyTorch · OpenCV · PyQt6 · FastAPI · Flask · Scikit-learn · Pandas · NumPy · Power BI · Supabase · PostgreSQL · Git · LLMs · RAG · Vector Database

Soft Skills: Agile Collaboration · Quantitative Reasoning · Impactful Communication · Technical Auditing · Critical Problem Solving · Project Management · Adaptability · Strategic Planning · Cross-functional Teamwork

Languages: Arabic (Native) · English (Fluent) · German (Beginner)